Biology-1(BIO103)

Lecture 01-02

Chapter 1. Introduction to Biology

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BIO103: Biology I Course Introduction

Lecture 1 & 2: Outlines Spring 2026

1. Course Objectives
2. Course content
3. Course Evaluation plan
4. Marks distribution-course evaluation
5. What is biology?
6. What is life?
7. Characteristics of living things

Biology I: Course Objectives

- This class is an **introduction to the science of biology** for non-majors.

- This is an introductory course.

- The fundamentals of **biology for the non-major**:
  - an overview of living organisms.
  - scientific inquiry,
  - biological chemistry,
  - cell structure and function,
  - DNA and genetics,
  - evolution and ecology,

Biology I: Course Content (Chapters)

1. Introduction to Biology: Characteristics of living things.
2. Chemistry of life
3. Biological Macromolecules
4. Central Dogma of Molecular Biology
5. Cell structure and function
6. Cellular Reproduction
7. Energy of Life: Metabolism
8. Human Physiology
9. Health and Diseases

Course Content and Evaluation plan

- Total Lectures: 24 (Approximate)
- Lecture 1 – 4: Quiz 1
- Lecture 5 – 8: Quiz 2
- Lecture 1 – 12: MID Exam
- Lecture 13 – 16: Quiz 3
- Lecture 17 – 20: Quiz 4
- Lecture 13 – 24: Final Exam
- **Best 3 quizzes will be counted.**
- **No make up quiz/midterm will be taken.**

Biology I: Course Evaluation plan

Marks Distribution

	Section	Marks %
1.	Attendance/Participation	10
2.	Quiz (**best 3 out of 4**) 3 X 10	30
3.	Mid Term Examination (1 X 30)	30
4.	Final Examination	30
Total Marks		**100**
Note: Some modifications of mark distribution may be made later, depending Departmental/NSU directives.		

Origin of Life

Big Bang Theory: Origin of Earth

: Explosion from a pin sized to a Huge Universe.
: Among all Planets, Earth only contains life.

Early earth/ earth without life

: High temperature, poisonous gases, no free oxygen/ water.
: Life evolved from chemical reactions and gradual changes.

Evolution of life

: Life begins when the cell forms.
: Single to multi-cellular organisms.
: Survival of the fittest.

What is Biology

- The word biology comes from the **Greek word bios**, which **means life**, and **logos, which means thought**.
- Thus, biology is the **science that deals with the study of life.**
- Biology = **the study of life**
- An attempt to understand, explain, integrate and describe the world of living things.

Approaches taken to study Biology

Taxonomy	The classification of organisms
Morphology	The study of the external form and structure of organisms
Anatomy	The study of the internal structure of organisms
Physiology	The study of the function in living system
Cytology	The study of cells
Ecology	The study of the relationship of organisms to their environment
Genetics	The study of inheritance
Pathology	The study of diseases

What is Life

- Life is the **condition** that distinguishes animals and plants from inorganic matter, including the **capacity for growth, reproduction, functional activity, and continual change preceding death.**

- Defined by **a set of biological processes** such as **organization (cells), metabolism, growth, adaptation, response to stimuli, and reproduction**

What is Life

- Life is d**efined by the capacity for growth, reaction to stimuli, metabolism, energy transformation, and reproduction**.

- **Various forms of life exist**, such as plants, animals, fungi, protists, archaea, and bacteria.

- Biology is the science that studies life.

- The **gene is the unit of heredity**, whereas the **cell is the structural and functional unit of life**.

Group of organisms being studied

1. Botany : The study of plants.
2. Zoology : The study of animals.
3. Microbiology : The study of microscopic organisms.
4. Bacteriology : The study of bacteria.
5. Virology : The study of viruses.
6. Mycology : The study of fungi.
7. Entomology : The study of insects.
8. Ornithology : The study of birds.

What is Life

- **Living organisms:**
  - Highly organized and complex.
  - Are composed of one or more cells.
  - Contain a blueprint of their characteristics.
  - Acquire and use energy.
  - Carry out and control numerous reactions.

What is Life

- **Living organisms:**
  - Grow.
  - Maintain constant internal environment.
  - Produce offspring.
  - Respond to environmental changes.
  - May evolve.

Characteristics of living things

There are nine characteristics of living things.

- The need for energy
- Movement
- Cellular structure and organization
- Growth and development
- Maintenance and repair
- Reproduction
- Response to stimuli
- Variation and adaptation
- Metabolism

Important Characteristics of Life

1. Living things are made up cells.
2. Living things reproduce.
3. Living things are based on a universal genetic code (DNA).
4. Living things grow and develop

Important Characteristics of Life

5. Living things obtain and use materials and energy.

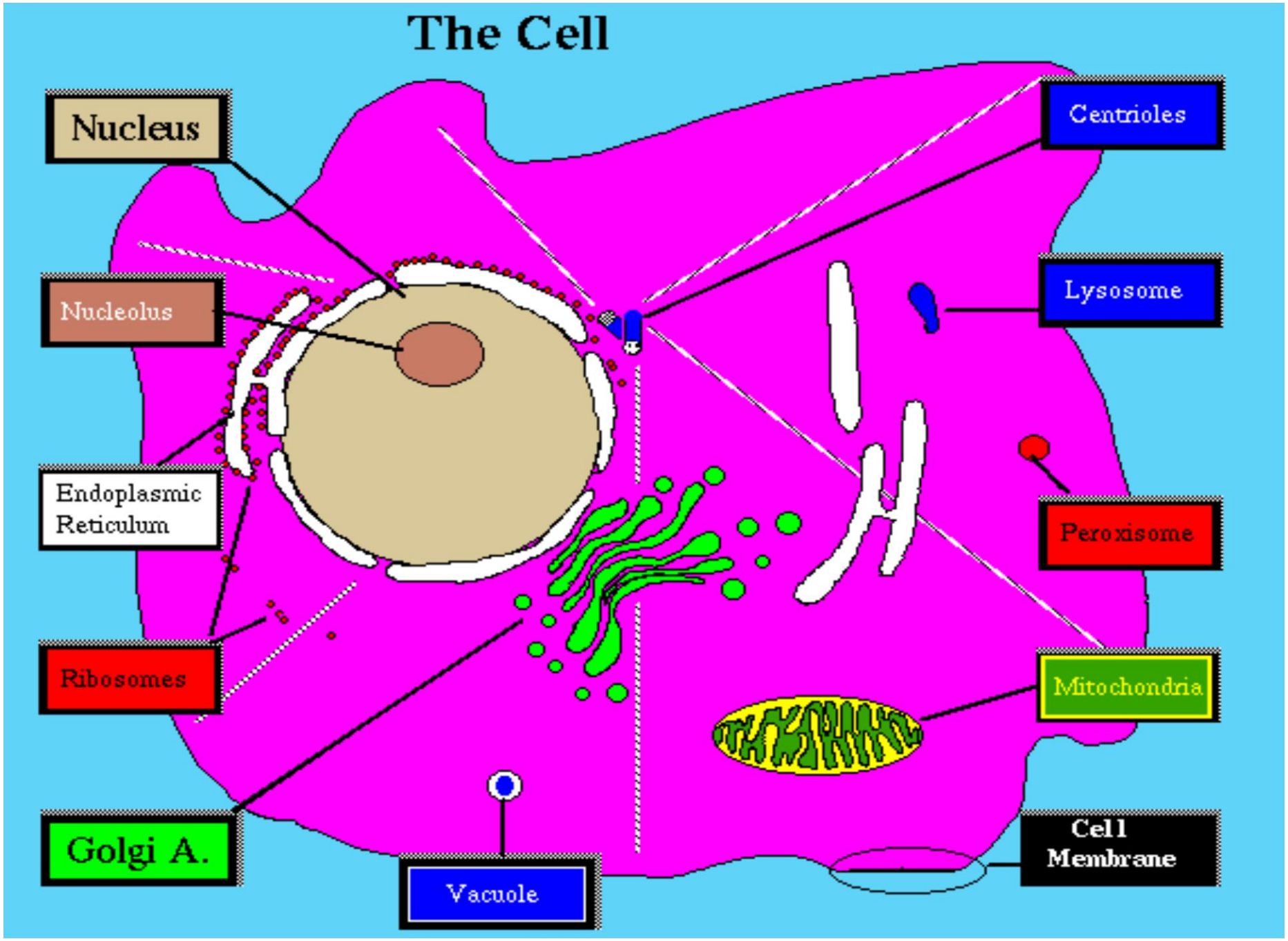
6. Living things respond to their environment.

7. Living things maintain a stable internal environment. (homeostasis)

8. Taken as a group, living things change over time.

Life: 1—Made up of cells

- Cell - every organism’s smallest unit of life.
- Unicellular - organisms are organisms made of only one cell.
- Living things that are made of two or more cells are called multicellular organisms.


The Cell
Nucleus
Centrioles
Nucleolus
Lysosome
Endoplasmic Reticulum
Peroxisome
Ribosomes
Mitochondria
Golgi A.
Vacuole
Cell Membrane

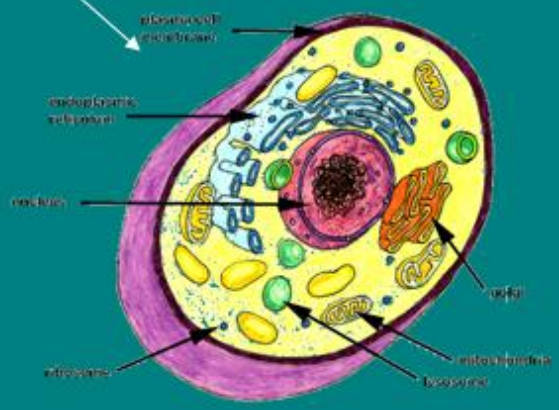
Life: 1—Made up of cells

They can be either unicellular or multicellular.

***Unicellular = made up of ONE cell

Picture of a unicellular organism.

Picture of a Cell



***Multi-cellular = made up of MORE than one cell

Picture of multicellular organisms

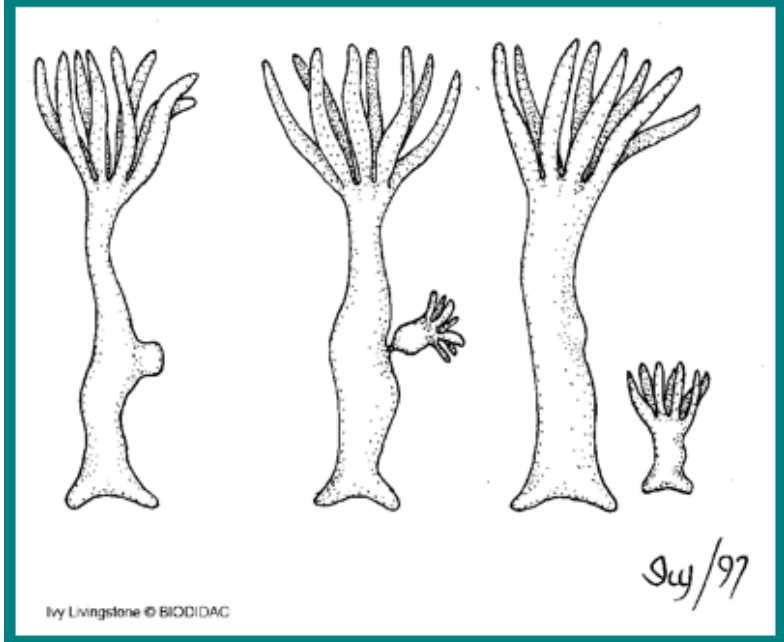
Life: 2—Reproduction

- Reproduction is the process by which one organism makes one or more new organisms.
- Some organisms can reproduce by dividing and becoming two new organisms. (Asexual)
- Some organisms must have a mate to reproduce (Sexual), but others can reproduce without a mate.

Life: 2—Reproduction

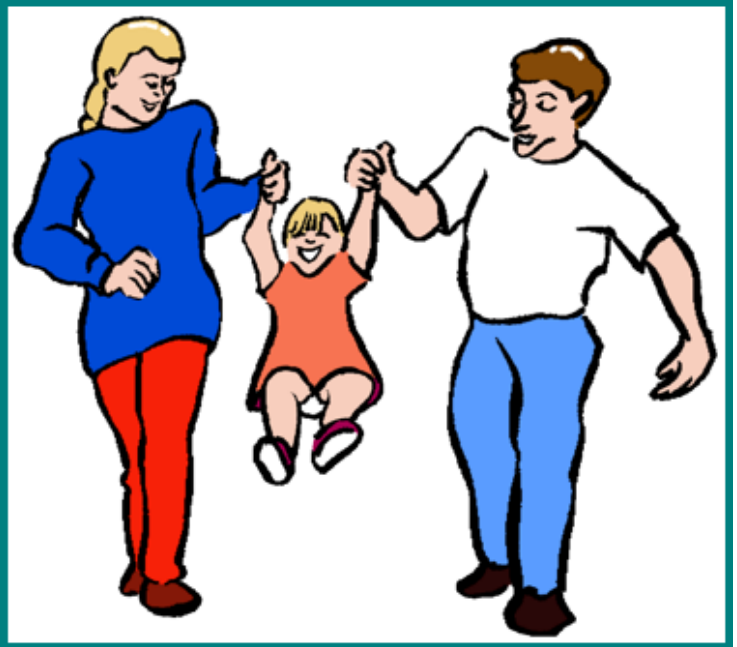
There are 2 types of reproduction:

1. Asexual



***<u>asexual</u> = reproduction that requires ONE parent

2. Sexual



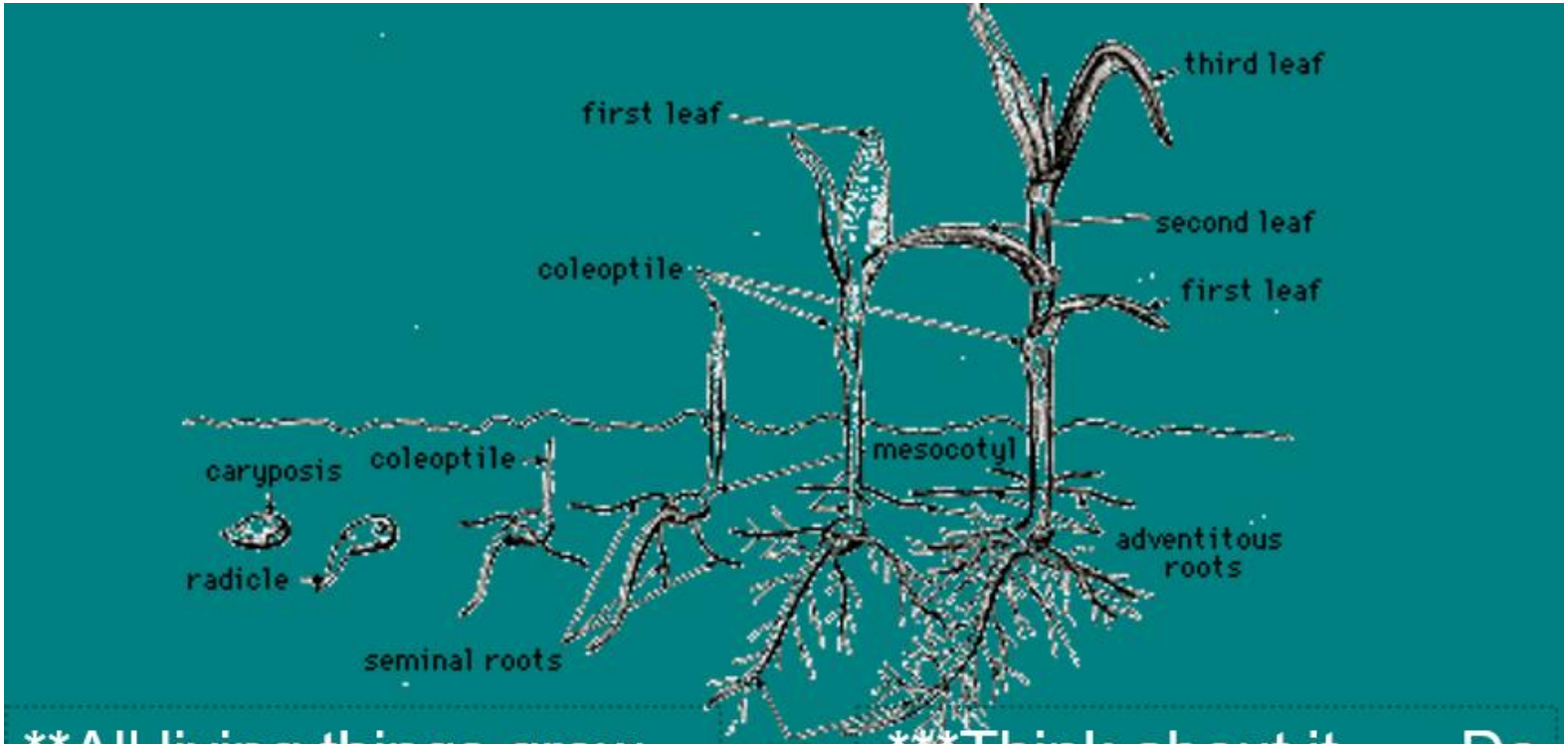
***<u>sexual</u> = reproduction that requires TWO parents

Life: 3—Growth and Development

- Changes that occur in an organism during its lifetime are called development.
- In multicellular organisms, development happens as cells become specialized into different cell types.
- Some organisms have dramatic developmental changes, such as a tadpole becoming a frog.

Life: 3—Growth and Development

Life: 3—Growth and Development



**All living things grow and develop during at least part of their life.

***Think about it . . . Do you look the same as when you were born????

Life: 4—Based on genetic code

- DNA is considered a universal genetic code because every known living organism has genes made of DNA.
- Bacteria, fungi, cats, plants, and you: every organism uses DNA to store genetic information.
- All organisms also use DNA to transcribe RNA, and then they translate that RNA into proteins.

Life: 4—Based on genetic code

DNA = DNA is the ***genetic code found in every cell. DNA is ***inherited*** from parents and controls your ***traits***.

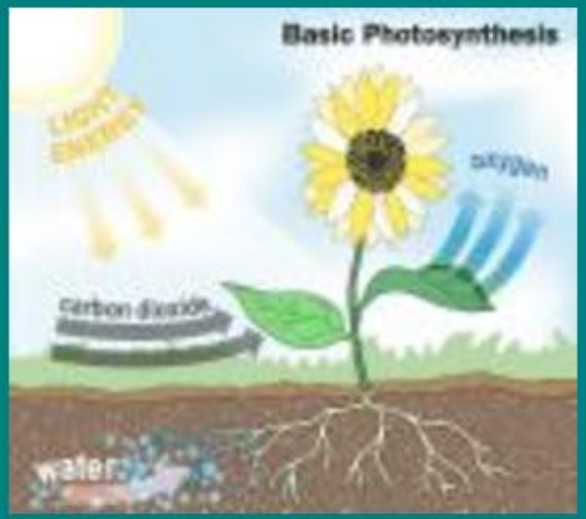
Life: 5—Need for Materials and energy

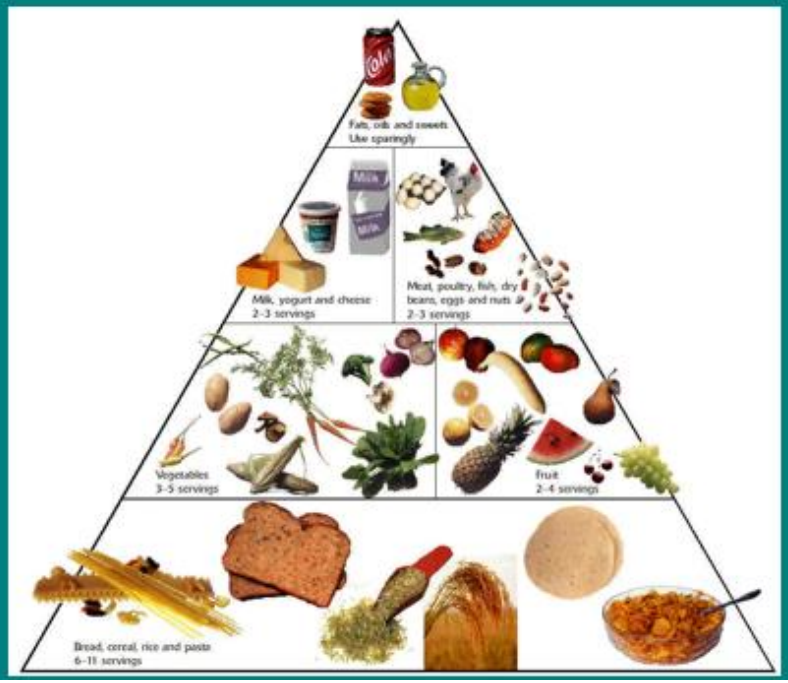
- Cells continuously use energy to transport substances, make new cells, and perform chemical reactions.
- For most organisms, energy for life comes from the Sun.
- The Sun's energy can be passed from one organism to another.

Life: 5—Need for Materials and energy

**Living things need a constant supply of materials and energy.

**Metabolism = the process of using those materials.





Life: 5—Need for Materials and energy

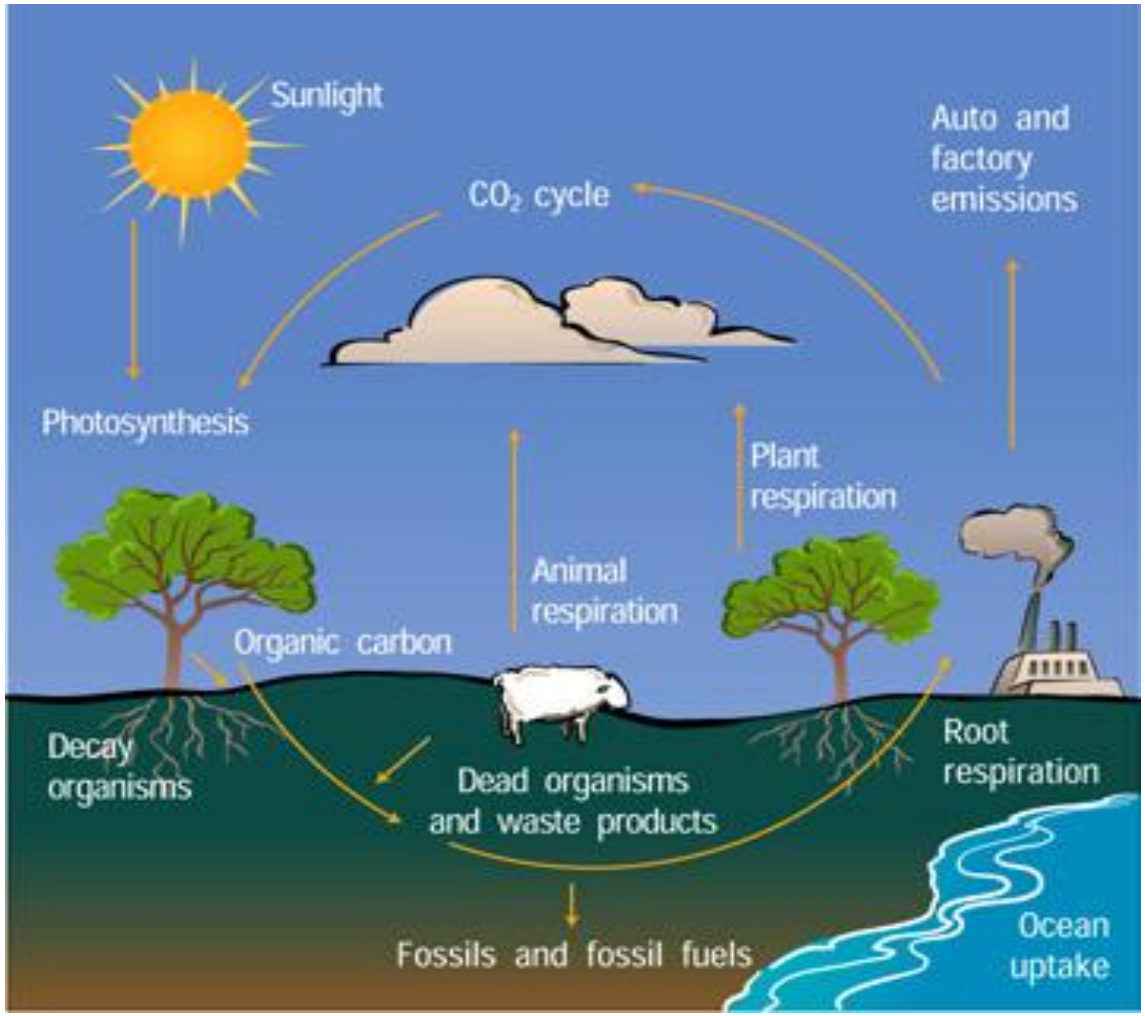


Figure: The Energy Cycle

Life: 6—Response to stimuli/environment

- All living things respond to changes in the environment called stimuli. There are two types of stimuli:

- Internal stimuli are changes **within** an organism.

- Common examples include hunger, thirst, pain, emotional states like fear or happiness, and physiological changes such as fever, low blood sugar.

- External stimuli are changes in an organism's **environment.**

- Examples include light, sound, temperature changes, odors. These, such as seeing a predator or feeling cold, allow organisms to adapt to their surroundings

Life: 7—Homeostasis (maintaining internal balance)

- An organism's ability to maintain steady internal conditions when outside conditions change.
- Homeostasis: Maintain a constant stable internal environment suitable for life.
- Maintaining homeostasis ensures that cells and the organism can function normally and remain healthy.
- **When an organisms is unable to maintain homeostasis it dies**
- Metabolism provides the raw materials and energy to maintain homeostasis

Life: 7—Homeostasis

Despite *changing* conditions *around* them, organism must keep their **internal** conditions **stable/balanced**.

<u>Homeostasis**</u> =
the **process** of keeping the **internal** environment **stable/balance/constant**

Life: 8— Metabolism

- **Metabolism** is the sum of all the chemical reactions that occur in the organism.

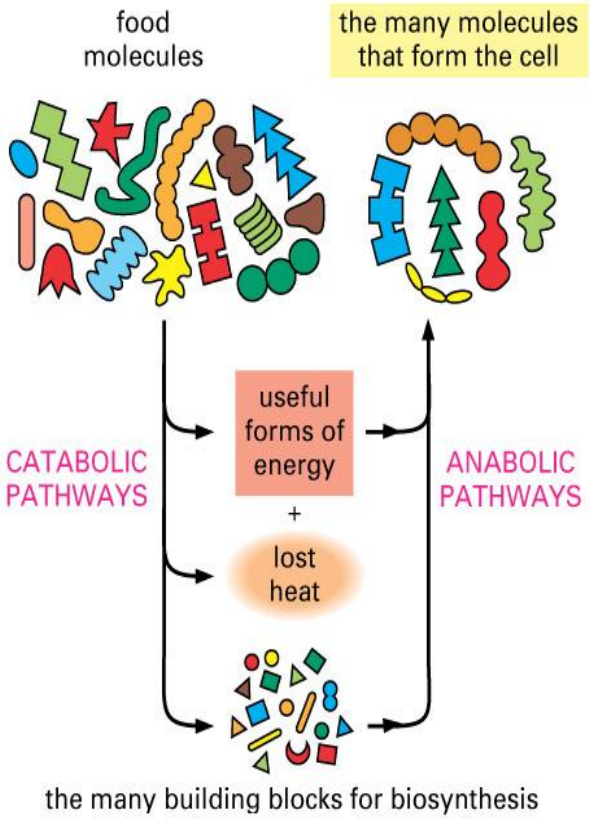
  - Ingestion
  - Digestion
  - Assimilation
  - Respiration
  - Excretion

Life: 8— Metabolism

Metabolism has two distinct phases--

1. Anabolism: Building up

2. Catabolism: Breaking down



Important Characteristics of Life

Characteristics of Life	
Characteristic	**Definition**
Organization	Living things have specialized structures with specialized functions. Living things with more than one cell have a greater level of organization because groups of cells function together.
Growth and development	Living things grow by increasing cell size and/or increasing cell number. Multicellular organisms develop as cells develop specialized functions.
Reproduction	Living things make more living things through the process of reproduction.

Characteristics of Life	
Characteristic	**Definition**
Response to stimuli	Living things adjust and respond to changes in their internal and external environments.
Homeostasis	Living things maintain a stable internal environment.
Use of energy	Living things use energy for all the processes they perform. Living things get energy by making their own food, eating food, or absorbing food.

Themes of Biology

- **Hierarchical organization of life**

  -- structural make up, from the smallest, simple to largest complex

- **Cell theory**

  -- all organisms are made of cells, central idea in all studies of biology

- **Heredity**

  -- Biological information is inherited from parents in one generation by the offspring in the next.

- **Evolution**

  -- the modification of populations over time

Themes of Biology

- **Regulation**
  -- To survive and reproduce, all forms of life must regulate their internal, and sometimes external, environment.
- **Structure and function**
  -- Correlations between the structure of a biological object, and its function.
- **Environmental interactions**
  -- Individuals interact with other organisms of their own species and those of the greater community.
- **Energy flow**
  -- Energy flows through a food chain, cycles through an ecosystem, or is converted to different forms within the cells of an organism.

So…
Biology: Redefined

- Biology was first defined as the science that deals with the study of life

- However, as we learn more, we see that biology involves many other things. It is also a study of all those things that affect life.

- Thus, the following is a more accurate definition of biology:

Biology is the study of living things and the things that were once alive, together with the matter and energy that surround them.

Questions?